Factory Reallocation and Shipping Optimization: A Decision-Support Framework for Product-to-Factory Assignment at Nassau Candy Distributor

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Abstract

A confectionery distributor must decide which factory should supply which product, because that decision shapes shipping distance, delivery cost and the profit retained on each sale. This paper presents a decision-support framework for the Nassau Candy Distributor, built on a transaction dataset of 10,194 rows and 18 columns. The framework has four layers: a data-quality audit, a profitability analysis by region, division and product, a geospatial layer that estimates great-circle distance between candidate factories and customers, and a transparent what-if simulator that compares a baseline product-to-factory assignment with alternatives. A formal assignment model with eligibility and capacity constraints is specified for the next phase. The audit produced the study's most important result: order dates fall in 2024–2025 while ship dates fall in 2026–2030, giving implied lead times of roughly 904–1,642 days. Lead-time modelling was therefore stopped rather than run on implausible data. The inspected schema also contains no observed factory, freight charge or capacity field, so every reallocation output is reported as a scenario estimate, not as realised savings. The paper contributes a reproducible pipeline, an explicit cost and distance model, a validation protocol, and a precise list of the data needed to turn the prototype into a defensible optimiser.

Keywords: decision intelligence, supply chain analytics, facility allocation, shipping optimisation, profitability analysis, data quality, Streamlit

1. Introduction

Distribution firms usually know their sales well and their fulfilment economics poorly. Revenue and gross profit are recorded per order, but the cost of moving a product from a particular factory to a particular customer is rarely visible in the same table. As a result,

decisions about where a product should be made or shipped from are often made by habit rather than by evidence.

This project asks how a small analytics team, working from a transaction file alone, can responsibly support that decision. The answer developed here is a staged framework. It describes what the data can already prove, tests what it cannot, and defines an optimisation model that can be run once the missing inputs are supplied.

1.1 Problem statement

Nassau Candy sells multiple products across several divisions and regions. Management wants to know whether reassigning products to different factories could reduce shipping distance and cost without harming profit or service. The available data describes sales, units, cost, gross profit, customer geography, shipping mode and dates. It does not record which factory fulfilled an order, what freight was charged, or how much each factory can produce.

1.2 Research questions

- **RQ1.** Is the transaction data reliable enough to support service-level and lead-time analysis?
- **RQ2.** Where are revenue and gross profit concentrated across regions, divisions and products?
- **RQ3.** How can customer-to-factory distance be estimated transparently from available geography?
- **RQ4.** How should alternative product-to-factory assignments be compared fairly against a baseline?
- **RQ5.** What additional evidence is required before a reallocation can be recommended to management?

1.3 Contributions

1. A reproducible workflow that moves from raw CSV to audited data, profitability analysis, distance estimation and scenario comparison.
2. A documented finding that the order and ship dates fail plausibility checks, together with a rule for handling it without altering the data.
3. An explicit cost-and-distance model and a constrained assignment formulation ready for use when inputs are validated.
4. A Streamlit dashboard that presents results with visible assumption warnings.

5. A validation and sensitivity protocol that separates observed fact from modelled estimate.

2. Background and Related Work

Facility location and product-to-plant allocation have a long history in operations research. Weber's classic location problem seeks a site that minimises weighted distance to demand points (Weber, 1909). Geoffrion and Graves (1974) extended this to multi-commodity distribution design, where products, plants and customers are decided together subject to capacity. Textbook treatments of supply-chain design (Chopra and Meindl, 2019) stress that transportation cost, service level and capacity must be traded off jointly, not in isolation.

Two ideas from this literature shape the present work. First, assignment decisions only make sense under capacity and eligibility constraints, since a nearer factory that cannot make a product is not a real option. Second, a model is only as credible as its inputs, so the data-quality audit is a first-class stage and not a preliminary chore (Wickham, 2014).

On method, the project follows a design-science approach (Hevner et al., 2004): it builds an artefact, a decision-support tool, and evaluates it against stated criteria while being explicit about the problem context and its limits.

3. Data and Quality Audit

3.1 Dataset

The dataset contains 10,194 transaction rows and 18 columns. The columns fall into five groups.

Group	Fields	Use in this study
Transaction	Row ID, Order ID	Record tracking, order grouping
Dates and shipping	Order Date, Ship Date, Ship Mode	Date audit, descriptive mode analysis
Customer geography	Country/Region, City, State/Province, Postal Code, Region	Regional summaries, distance estimation

Group	Fields	Use in this study
Product and organisation	Division, Product ID, Product Name	Product and division aggregation, candidate mapping
Commercial measures	Sales, Units, Gross Profit, Cost	Profitability and scenario inputs

Two supporting files in the repository hold the study's assumptions as explicit, reviewable inputs: a candidate factory-location file (name, latitude, longitude) and a proposed product-to-factory mapping file. Both are labelled assumptions until an authoritative source confirms them.

3.2 Preparation rules

- Keep the raw CSV unchanged and work on a copy.
- Check types, row counts, missing values, duplicates and numeric conversion failures. In the dashboard, Sales, Gross Profit, Cost and Units are coerced to numeric with errors surfaced as missing values rather than hidden.
- Parse dates using the source's day-month-year format, then inspect ranges and order-to-ship differences.
- Compute margin from group totals, and only where total sales are positive.
- Never overwrite a source value to make a result look plausible.

3.3 Critical finding: implausible dates (RQ1)

The audit found Order Date values spanning 2024–2025 and Ship Date values spanning 2026–2030. Implied order-to-ship intervals ranged from about 904 to 1,642 days, with a mean near 1,320.84 days and a median near 1,274 days. The year component of the Order ID also did not match the parsed Order Date year. A confectionery order does not take three to four years to ship, so these fields are either synthetic, transformed, or incorrectly assembled. The file alone cannot reveal which.

The decision taken was to **stop lead-time modelling**. The project does not shift dates, infer replacements from Order IDs, or drop rows to force plausibility, because each of those would introduce an unverifiable assumption into every downstream number. A corrected source file or written clarification from the data owner is the required remedy. This is also the answer to RQ1: the data supports commercial analysis but not delivery-time analysis.

4. Methodology

4.1 Pipeline overview

The workflow runs in five stages, each implemented as a separate notebook so that results can be traced and re-run independently.

1. **Date validation.** Range checks, interval statistics, Order ID consistency.
2. **Sales and profitability analysis.** Aggregation by region, division and product.
3. **Factory-location analysis.** Candidate factory coordinates and the proposed product mapping.
4. **Shipping-distance analysis.** Great-circle distance from each candidate factory to each customer location.
5. **Reallocation simulation.** Baseline versus alternative assignments under stated assumptions.

4.2 Profitability measures (RQ2)

For any group *g* of transactions:

- Total sales $S_g = \Sigma$ Sales
- Total gross profit $G_g = \Sigma$ Gross Profit
- Group margin $M_g = 100 \times G_g / S_g$, defined when $S_g > 0$

Margin is computed from group totals instead of averaging row-level margins, so that large orders are weighted correctly. Products are ranked on both gross-profit contribution and margin, since a high-margin item with small volume can contribute less profit than a modest-margin bestseller. Shipping mode is profiled by order count, sales, units and gross profit, but is treated as descriptive only: mode alone does not establish cost or speed.

4.3 Distance estimation (RQ3)

Where customer and factory coordinates are valid, the great-circle distance between points (φ_1, λ_1) and (φ_2, λ_2) is estimated with the haversine formula (Sinnott, 1984):

$$d = 2R \cdot \arcsin\left(\sqrt{ \sin^2(\Delta\varphi/2) + \cos\varphi_1 \cdot \cos\varphi_2 \cdot \sin^2(\Delta\lambda/2) } \right), \text{ with } R \approx 6{,}371 \text{ km}.$$

This is a straight-line proxy. It is not road distance, carrier routing distance or freight cost, and the geodesic error of the spherical assumption is small relative to that gap (Karney, 2013). Missing, ambiguous or implausible geocodes are flagged and excluded from

distance statistics, never silently imputed.

4.4 Estimated shipping cost

Because no freight charge is observed, shipping cost is modelled and every parameter is disclosed. For an order *o* of u_o units shipped from factory *f* to customer location *l_o*:

ĉ(o, f) = u_o · (α_m + β_m · d(f, l_o))

where m is the ship mode, α_m a fixed handling cost per unit and β_m a distance-linked rate per unit-kilometre. The parameters α_m and β_m are scenario inputs, to be calibrated against freight invoices or a published carrier tariff. Until then, results are reported as sensitivity ranges, not point savings.

4.5 Scenario comparison (RQ4)

A scenario is a complete product-to-factory mapping together with its cost parameters. Each scenario is evaluated on the **same set of orders** as the baseline, so that differences arise from the assignment and not from different data. For each scenario the simulator reports total estimated shipping cost, mean and distance-weighted shipping distance, and freight-adjusted gross profit G − ĉ. Changes are expressed relative to the baseline, and the baseline itself is flagged as unverified until the true current assignment is confirmed.

4.6 Constrained assignment model (future phase)

Let P be the set of products, F the candidate factories, and O_p the orders of product p. Define the assignment cost of product p at factory f as

C_pf = Σ over o in O_p of ĉ(o, f).

Let x_pf = 1 if product p is assigned to factory f, and 0 otherwise. The model is:

Minimise Σ_p Σ_f C_pf · x_pf

subject to

- Σ_f x_pf = 1 for every product p (each product has one source)
- x_pf ⩽ e_pf for all p, f (assign only where the factory can make the product)
- Σ_p q_p · x_pf ⩽ K_f for every factory f (capacity)
- d̄_p(f) ⩽ D_max for assigned pairs (a service-distance cap), where service data allow
- x_pf ∈ {0, 1}

Here e_pf is an eligibility flag, q_p the production volume of product p, K_f the capacity of factory f, and D_max a stakeholder-set limit. This is a generalised assignment problem. At the scale of a candy catalogue and a handful of factories it is solved exactly by any standard integer-programming solver. The formulation is specified now so that the work needed is purely a matter of supplying validated inputs. It has **not** been solved on real data in this paper.

5. System Implementation

The repository is organised as notebooks for analysis, CSV files for inputs, and a Streamlit application as the interface.

Component	File	Role
Date validation	05_Date_Validation.ipynb	Range and interval checks
Profitability	06_Sales_Profitability_Analysis.ipynb	Region, division, product summaries
Factory locations	07_Factory_Location_Analysis.ipynb	Candidate sites and mapping
Distance	08_Shipping_Distance_Analysis.ipynb	Customer-to-factory distance
Simulation	09_Factory_Reallocation_Simulation.ipynb	Baseline vs alternative scenarios
Inputs	Nassau Candy Distributor.csv, factory_locations.csv, proposed_product_factory_mapping.csv	Data and explicit assumptions
Interface	appp.py, requirements.txt	Dashboard and dependencies

The dashboard has four tabs. **Overview** shows total sales, gross profit, cost, units and aggregate margin with region filters, plus regional bar charts. **Product Analysis** ranks products by gross profit and lists sales, profit and units. **Factory Assignments** displays the proposed mapping and candidate locations beneath a standing warning that the assignments are unvalidated. **Data Explorer** shows the filtered table with a CSV

download. Metrics are recomputed from the filtered data on each change. The interface closes with a note that historical sales and profit do not prove that reallocation will reduce cost, and that date fields still require validation.

A design principle is that warnings live next to the numbers they qualify, so a reviewer cannot read a chart without also seeing its limits.

6. Findings to Date

Three conclusions are supported by the work completed.

Finding 1 – Commercial analysis is feasible. The dataset carries the dimensions needed for meaningful sales and gross-profit summaries by region, division and product, and these views are live in the dashboard.

Finding 2 – Date-based analysis is not currently valid. The 904–1,642-day intervals and mismatched Order ID years make shipment lead-time modelling unsound. The correct response was to halt modelling and request clarification.

Finding 3 – The schema cannot by itself prove an optimal reallocation. There is no observed factory, freight cost or capacity field. Any claim of optimality would rest on assumptions the data cannot check.

The figures for the headline results of Finding 1 should be taken directly from the project notebooks and inserted below before submission:

Measure	Value
Total sales	[insert from notebook 06]
Total gross profit	[insert from notebook 06]
Overall gross-profit margin	[insert from notebook 06]
Highest-sales region / highest-profit region	[insert from notebook 06]
Top three products by gross profit	[insert from notebook 06]
Median customer-to-assigned-factory distance (km)	[insert from notebook 08]

7. Evaluation Protocol

Evaluation is organised in layers so that each kind of error is caught at the layer where it arises.

Layer	Check	Pass criterion
Data validation	Reconcile dates, IDs and gross-profit definitions with source documentation	Dates plausible and documented
Descriptive accuracy	Recompute group totals and margins independently in a spreadsheet or SQL	Match to rounding
Distance validation	Sample-check geocodes; compare haversine with road distance for a subset	Errors understood and disclosed
Cost calibration	Fit α_m and β_m to freight invoices or a tariff	Residuals within agreed tolerance
Scenario evaluation	Baseline and alternatives on the same orders	Reported with service constraints
Sensitivity	Vary freight rates, distance caps and capacity	Recommendation stable across range
Stakeholder review	Operations confirms eligibility and practicality	Written sign-off

A recommendation is considered robust only if the preferred assignment remains the best, or within an agreed margin of the best, across the sensitivity range, for example freight rates varied by ±25%.

8. Discussion

What the work establishes. It establishes that the commercial data is usable, that the date data is not, and what a defensible optimisation would require. Many analytics projects of this kind present a ranked list of factory moves with savings figures. Here

those figures would be fabrications, because the baseline, the freight cost and the capacity are all unobserved. Declining to report them is a methodological result in its own right.

Managerial implications. Management can already use the profitability views to see where gross profit is concentrated and which products justify attention. For reallocation, the practical next step is not further modelling but a short data request: current factory per product, freight invoices for a sample period, factory capacities, and corrected dates.

Where reallocation is most likely to matter. The distance model suggests value is greatest for high-volume, high-margin products sold in regions far from their current source, since shipping cost is a larger share of the profit there. Whether this holds is an empirical question that the validated model, not this paper, must answer.

9. Limitations and Threats to Validity

- **Unobserved baseline.** No factory-assignment field is present, so the current state is itself assumed.
- **No freight data.** Cost is modelled, and savings are therefore estimates that depend on α_m and β_m.
- **Date integrity.** Delivery-time conclusions are suspended pending a corrected source.
- **Distance proxy.** Great-circle distance ignores road networks, carrier hubs and cross-border effects.
- **Constraints not yet modelled.** Capacity, inventory, production capability and customer commitments are outside the current data.
- **Mapping provenance.** The proposed product-to-factory mapping is an assumption until confirmed.
- **External validity.** Results describe one distributor's dataset and should not be generalised without replication.

10. Ethical and Reproducibility Considerations

Raw data is preserved, transformations are documented, and assumptions are visible in code and in the interface. Customer location fields can be sensitive, so they should be anonymised or removed before any public release, and the dataset's licence and internship confidentiality rules should be checked before the source file is shared. Dependencies are listed in requirements.txt. Any future predictive model must use a justified train-test split and checks for data leakage.

11. Conclusion and Future Work

This paper presented a decision-support framework for product-to-factory assignment at Nassau Candy Distributor that combines a data audit, profitability analysis, geospatial distance estimation, a transparent scenario simulator and a dashboard. Its central methodological choice was to halt lead-time modelling when the dates proved implausible, and to report reallocation outputs as scenarios and not as savings.

Future work proceeds in four steps: (1) obtain a corrected dataset and confirm current factory assignments; (2) add freight invoices or a documented tariff and calibrate the cost model; (3) collect eligibility and capacity data and solve the constrained assignment model in Section 4.6 with sensitivity analysis; and (4) review the result with an operations stakeholder and, if approved, pilot a limited reassignment while measuring realised cost and service change against the baseline.

Appendix A – Reproducibility Checklist

- Keep the raw dataset unchanged and record its source and version.
- Run date-quality checks before computing any lead time.
- Document every assumption in the product-factory mapping file.
- Validate factory coordinates and customer geocodes.
- Separate descriptive facts from scenario estimates in charts and text.
- Pin dependency versions in requirements.txt and add usage steps to README.md.
- Do not publish restricted data, personal information or confidential internship material.

Appendix B – Notation

Symbol	Meaning
p, f, o	Product, factory, order
S_g, G_g, M_g	Group sales, gross profit, margin
d(f, l)	Great-circle distance from factory f to location l
α_m, β_m	Fixed and distance-linked cost parameters for ship mode m
ĉ(o, f)	Estimated shipping cost of order o from factory f

Symbol	Meaning
C_pf	Assignment cost of product p at factory f
x_pf	1 if product p is assigned to factory f
e_pf, K_f, q_p, D_max	Eligibility, capacity, product volume, distance cap

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